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My Courses > UE23EC342BB3 : Architectures for Hardware Acceleration > Hardware Design for finite field inverse

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★ Applicable to all classes of this unit

Max Marks: 12

1) Which of the statement is true?

- ☒ Power supply noise is bad for PUF and good for TRNG
- ☐ Temperature variation is bad for PUF and good for TRNG
- ☐ Power supply noise is good for PUF and bad for TRNG
- ☐ Aging is bad for PUF and good for TRNG

2) Phit in NoC represents the

- ☒ the set of bits that are sent in a single clock cycle
- ☐ the set of bits that are sent in two clock cycles.
- ☐ the set of bits that are sent in bytes/bits in a single clock cycle

 Entrance Exam☐ the set of bits that are sent as packets. Announcements My Profile User Vehicle Backlog
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3) The Physical Unclonable Function (PUF) which support limited/small number of quality challenge response pairs is called

☐ Strong PUF☐ Limited PUF☐ Obfuscated PUF☒ Weak PUF

4) A situation where we have a set of flits (in NoC) that are not able to make progress because of circular wait is known as

☒ deadlock☐ Livelock☐ Starvation☐ drought

5) The Physical Unclonable Function generate the CRPs by exploiting the

☐ Power supply noise☐ Gaussian noise due to electromagnetics☒ Process variations occurring during manufacturing

☐ Changes in temperature

6) A Physical Unclonable Function (PUF) which measures the distinctive challenge-response pair (CRP) generation quality, that is, distinguishability, of a PUF with respect to other instances is called

☒ Uniqueness

☐ Responsiveness

☐ Reliability

☐ Uniformity

7) In the context of the Network on Chips (NoC) the packet is divided into several flow control digits, referred as

☒ Flits

☐ Phits

☐ Packet

☐ address

8) Successful modelling attacks on PUF require an access to

☒ Large number of PUF devices to collect sufficient number of challenge response pairs.

☐ Process related details of PUF

- ☐ Design related details of PUF
 - ☐ Temperature related data of the PUF
-

9) Butterfly PUF is designed by exploiting the

- ☒ Metastable properties in the FPGA matrix
 - ☐ Jitter and skew sensitivities in the FPGA matrix
 - ☐ Clocking circuit properties in FPGA matrix
 - ☐ Macro's design and interconnect properties in FPGA matrix
-

10) Ring Oscillator PUFs fail to generate a reliable response if

- ☐ a frequency crossover happens in ring oscillator after possible frequency degradation due to process variation.
 - ☐ a frequency crossover happens in ring oscillator after possible frequency degradation due to temperature variations
 - ☐ a frequency crossover happens in ring oscillators after possible frequency degradation due to modelling attacks
 - ☒ a frequency crossover happens in ring oscillators after possible frequency degradation due to aging.
-

11) AES supports the

- ☒ 128 bit key with 10 rounds
- ☐ 128 bit key with 12 rounds
- ☐ 128 bit key with 14 rounds

☐ 192 bit with 10 rounds

12) The ?Mix columns? is involved in AES as below

- ☐ All rounds will have Mix Columns
- ☒ All rounds will have Mix Columns except last round
- ☐ All rounds will have Mix Columns except first round
- ☐ Only one round randomly will have Mix Column

Your score is 0

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AES algorithm ▶

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